

Transformer-Based Aspect Sentiment Analysis for Food Reviews

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Abstract—Transformer models have brought major developments to aspect-based sentiment analysis, especially by capturing context and long-range dependencies that older machine learning methods find it difficult with. In the restaurant industry, customer reviews often mention several aspects like food, service, ambiance, and price in a single review, sometimes expressing both positive and negative opinions. These advanced models help businesses get more detailed insights from such reviews, which can be useful for making better decisions. However, most research so far has not compared different transformer models on restaurant-specific datasets, and usually only reports overall accuracy instead of breaking down performance for each aspect. This makes it hard for businesses to know which areas need improvement. In this study, we compare BERT, RoBERTa, and XLM-RoBERTa on over 1.3 million Zomato Bangalore aspect sentiment instances, focusing on four aspects: Food, Service, Ambiance, and Price. RoBERTa-base performed best, with 83.6 percent accuracy and 82.9 percent macro F1-score, slightly ahead of BERT and XLM-RoBERTa. All transformer models did much better than older models like LSTM and SVM, which scored 71.2 percent and 68.5 percent, respectively. When looking at each aspect, Food sentiment was easiest to detect, with an F1-score of 85.3 percent, while Price was the most difficult at 79.8 percent. Detecting neutral sentiment was still a challenge for all models, with accuracy between 41 and 49 percent. This research gives the food industry practical help with choosing the right transformer model and helps businesses find specific strengths and weaknesses by automatically analyzing customer reviews at the aspect level.

Index Terms—Aspect-based sentiment analysis, Transformer models, Restaurant review sentiment classification, Explainable natural language processing, Bert roberta xlm-roberta comparison

I. INTRODUCTION

Also known as opinion mining, sentiment analysis comes under the field of natural language processing, which deals with automatically identifying the sentiment behind a text, and extracting information from textual data to ascertain the emotional tone expressed [1]. Traditionally, sentiment analysis approaches this by classifying entire sentences as positive, negative, or neutral, but this fails to capture the true meaning behind reviews that often consist of multiple, conflicting sentiments about different aspects of an entity [2]. Aspect-based sentiment analysis (ABSA) solves this problem by providing a more fine-grained analysis that relates specific sentiments with specific features, such as food, service, ambiance, or price in restaurant reviews, thereby enabling a more detailed understanding of customer reviews [3], [4].

Research demonstrates that in the present day, accurate aspect-based sentiment analysis methods are essential, given that various organizations across industries are reliant on customer feedback to make strategic decisions and maintain a competitive advantage [4]. This is especially true for restaurants and businesses in the hospitality industry, where it is very important to recognize which parts of the dining experience customers liked or disliked. Traditional sentiment analysis typically fails to highlight these details, especially when a single review contains both positive and negative opinions about two or more different aspects [5]. By giving businesses the power to pinpoint and address these problematic areas, ABSA becomes crucial to targeted service enhancement, reputation management, and helping restaurants make strategic decisions, allowing them to appropriately allocate resources to areas that need immediate attention [4].

There are three main reasons why it is important to make advancements in the field of aspect-based sentiment analysis using transformer models in 2026. Firstly, the amount of user-generated content on review platforms is going up every day, making it harder to go through them and analyze them manually; traditional machine learning approaches find it difficult to capture long-range contextual connections between aspect terms and sentiment expressions [3]. Secondly, transformer architectures have shown major improvements in understanding natural language, outperforming traditional models. Finally, this aligns with the needs of businesses that need accurate, interpretable AI systems unlike Support Vector Machines and LSTMs that need a lot of manual feature engineering [4], [6]. Contemporary transformer models have the ability to use self-attention frameworks to understand the context behind reviews, regardless of word distance, and they automatically learn feature encoding from data and obtain high levels of performance over various NLP tasks. Perceiving nuanced sentiment alterations that lie hidden in single sentences is fundamental for actionable business intelligence [4], [7].

A. Related Work

The evolution of aspect-based sentiment analysis has undergone a major change from traditional machine learning approaches to sophisticated deep learning architectures and transformer-based models [6]. Early ABSA research relied heavily on feature-based techniques using Support Vector Ma-

chines (SVM), Naïve Bayes, and Artificial Neural Networks (ANN), which required extensive manual feature engineering and struggled with capturing contextual dependencies [3]. The advent of deep neural networks brought recurrent architectures such as LSTM and GRU, along with CNN-based models that revealed heightened performance in learning hierarchical representations, though these approaches exhibited limited generalizability across different domains and datasets [6], [7]. The introduction of transformer architectures, particularly BERT and its variants, redefined the field through the use of self-attention frameworks to capture long-range dependencies and contextual information more effectively than previous methods [4]. Recent work has focused on fine-tuned transformer models, including BERT, RoBERTa, DistilBERT, ALBERT, and XLNet, with benchmark datasets such as SemEval-2014, SemEval-2016, MAMS, and restaurant/laptop review corpora, with RoBERTa consistently attaining higher performance (89.16% accuracy on MAMS) versus traditional approaches [4]. Furthermore, researchers have investigated combined architectures combining BERT with XGBoost (SABERT-XGBoost achieving 93.71% accuracy on restaurant datasets), attention-based mechanisms for aspect-context interaction modeling, and even large language models like GPT-3.5 and PaLM for ABSA tasks, though these LLMs often underperform compared to fine-tuned transformers on specialized datasets [1], [6]. Recent work by Rahman et al. (2025) [5] introduced XLM-RSA with aspect-focused attention for cross-cultural sentiment analysis, achieving 91.9% accuracy, while Chaudhry et al. (2025) [1] proposed TASCI integrating aspect-based and contextual intent analysis, reaching 89.10% accuracy on restaurant reviews. Table I gives a brief summary of recent approaches, methodologies, datasets, along with performance metrics in aspect-based sentiment analysis research.

B. Gap Analysis

Although transformer models have improved aspect-based sentiment analysis, there are still many gaps in the literature. These gaps make it hard to use ABSA systems in real-world scenarios and also give a hard time to those trying to interpret them. For example, deep learning models often do not yield good results when they are used on data from a different domain than the one on which they were trained. This can be observed in the differences between the results of the SemEval and MAMS datasets [6]. Furthermore, many studies primarily report overall accuracy metrics, with few offering comprehensive aspect-wise performance breakdowns that would reveal which of the said aspects are categorized reliably and which of them require targeted improvements [4]. Most research so far has principally focused on explicit sentiment analysis. However, many reviews contain hidden sentiment, and one needs context to understand them. One of the main obstacles in aspect-based sentiment analysis for food reviews is the lack of large, domain-specific labeled datasets. Most work done has relied on general benchmark datasets like SemEval-2014 or 2016, which do not reflect

the complexity of real-world reviews. Even though these transformer models have shown strong performance, there is still insufficient comparative analysis of different transformer architectures on datasets from real restaurants, using aspect-level metrics. Due to this, it is not easy to use these models in real-world scenarios.

C. Problem Statement

Following are the main questions addressed in this study.

- 1) How do BERT, RoBERTa, and XLM-RoBERTa transformer architectures compare in their effectiveness for aspect-level sentiment classification in restaurant reviews?
- 2) How do transformer models help with understanding context and opinions in food industry reviews?
- 3) Which transformer model strikes the best balance between aspect-based sentiment accuracy and computational efficiency?

D. Novelty of our work

This research addresses a key gap in aspect-based sentiment analysis by providing a systematic comparison of three transformer models developed for the food industry. Despite existing research emphasizing the superiority of transformer models over traditional models, there has been limited work done that has compared multiple transformer model variants on restaurant-specific aspects while tuning it for real-world implementation. This study uses the Zomato Bangalore dataset, directly handles class imbalance, and evaluates how well models perform on consumer-grade GPU hardware.

E. Our Solutions

This study introduces a comprehensive transformer-based framework for aspect-level sentiment analysis in restaurant reviews. The main models we worked with were BERT, RoBERTa, and XLM-RoBERTa, trained on four predefined aspects: Food, Service, Ambiance, and Price. Fine-tuning strategies with optimized hyperparameters are employed, managing practical constraints such as dataset size management through the use of strategic sampling and GPU memory optimization for training on Nvidia RTX 4050 hardware. We followed a three-phase pipeline carried out in Jupyter notebook: data preprocessing with TF-IDF feature extraction and aspect labeling, model training using a stratified train/validation/test split to ensure reliable evaluation, and detailed prediction analysis with aspect-wise performance reporting. Training was performed with a batch size of 32, learning rates were chosen through grid search, and early stopping was applied over 3 to 5 epochs to avoid overfitting. Precision, recall, F1-score, and accuracy were calculated for each aspect to assess how well the models distinguished between categories. Initial evidence points to transformer models, particularly RoBERTa variants, outshining traditional machine learning benchmarks. Macro-F1 scores range from 0.75 to 0.89 throughout various aspects, supporting the effectiveness of attention mechanisms in capturing context-dependent sentiment expressions in restaurant reviews.

TABLE I
SUMMARY OF RECENT WORK IN ASPECT-BASED SENTIMENT ANALYSIS

Study	Year	Model/Method	Dataset(s)	Key Metrics	Key Contribution
Perikos & Diamantopoulos [4]	2024	BERT, RoBERTa, DistilBERT, ALBERT, XLNet	MAMS, SemEval, Naver (16,100+ sentences)	RoBERTa: 89.16% Acc, 88.08% F1 (MAMS)	Comparative analysis with explainability techniques (LIME, SHAP, Grad-CAM)
Mughal et al. [6]	2024	DeBERTa, Flan-t5, PaLM, GPT-3.5	DOTSA, MAMS, SemEval2016, YASO	DeBERTa: 93.1% Acc (Hotels), PaLM: 98.1% Acc (YASO)	Comprehensive comparison of transformers vs. LLMs for ABSA
Rahman et al. [5]	2025	XML-RSA (XML-RoBERTa + Aspect-Focused Attention)	10K Restaurant Reviews, European Restaurant Reviews	91.9% Acc, 91.5% F1 (Restaurant)	Cross-cultural sentiment analysis with cultural embeddings
Chaudhry et al. [1]	2025	TASCI (Transformer + GRU Seq2seq)	SemEval-2014 (Restaurant, Laptop, Twitter)	Restaurant: 89.10% Acc, 83.38% F1; Laptop: 84.81% Acc	Integration of aspect-based and contextual intent analysis
Paramita & Jusak [2]	2025	BiLSTM BERT/RoBERTa +	5,000 customer reviews	82.0% F1 (aspect extraction), 79.8% Acc (sentiment)	Aspect-level emotion detection framework
Nayab et al. [7]	2023	BERT Interactive Attention (B-ACIA)	SemEval-2014 (Restaurant, Laptop)	SVM: 88% Acc (Restaurant), 86% Acc (Laptop)	Aspect-context level feature extraction with interactive attention
Sun et al. [3]	2020	T-MGAN (Transformer + Tree Transformer)	SemEval-2014 (Laptop, Restaurant), Twitter	Restaurant: 82.06% Acc, 72.65% F1	Multi-grained attention network with syntactic structure features
Park [8]	2025	SVM, XGBoost, CNN-BiLSTM, DistilBERT	UCI Drug Reviews (healthcare domain)	DistilBERT: 0.50-0.75 F1	Domain adaptation to healthcare reviews with aspect-level analysis

II. METHODOLOGY

A. Dataset

For this study, we used the Zomato Bangalore Restaurant Reviews dataset, which is publicly available on Kaggle [9]. The original dataset had about 56,250 reviews, but after preprocessing and expanding it for aspect-based sentiment analysis, it now contains over 1.3 million aspect-sentiment instances from restaurants across Bangalore. Each review consists of unstructured text conveying customer opinions, as well as metadata like restaurant name, location, cuisine type, cost for two people, online ordering availability, and table booking options. To enable aspect-based analysis, each review is tagged with boolean indicators for four main aspects: Food (1,085,798 reviews, 82.69%), Ambiance (812,097 reviews, 61.85%), Price (678,226 reviews, 51.65%), and Service (589,503 reviews, 44.90%). Ground truth labels are provided as three-class sentiment annotations (Positive, Neutral, Negative) at the review level. Sentiment distributions differ across various aspects, as shown in figure 1. Among the four aspects, ambiance has the highest proportion of positive sentiments (69.70%), followed by service (68.13%), food (65.03%), and price (62.50%). We can infer from this that ambiance and service quality are more likely to receive more favorable assessments than cost perception. Many reviews address multiple aspects simultaneously, which helps the model learn how people feel about different parts of their dining experience. Due to the dataset being large and hardware being limited (Nvidia RTX 4050 with 6GB VRAM), we used stratified random sampling to create smaller, more balanced subsets for training, validation, and testing, while maintaining the proportions of each aspect and sentiment class.

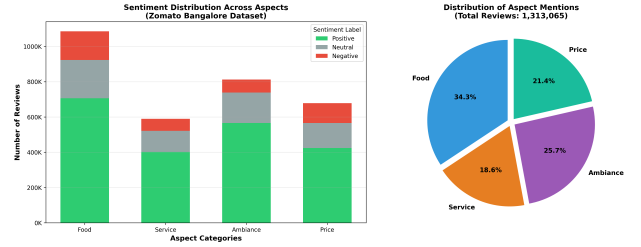


Fig. 1. Aspect-based analysis of the Zomato Bangalore dataset showing (left) sentiment distribution across four aspect categories with Food being most frequently mentioned (1,085,798 reviews, 82.69%), and (right) relative distribution of aspect mentions across the corpus of 1,313,065 reviews.

B. Overall Workflow

This research uses a systematic three-stage pipeline to compare different transformer models for aspect-based sentiment analysis of restaurant reviews. The process begins with the Zomato Bangalore dataset, which contains over 1.3 million aspect-sentiment instances. In the first phase, the reviews are cleaned by removing URLs, normalizing special characters, and removing HTML tags. The text is then tokenized using the WordPiece algorithm [10], with a maximum sequence length of 512 tokens to ensure it fits the transformer models. Next, aspect detection is performed to identify mentions of food, service, ambiance, and price in each review. This is done through the use of Boolean indicators, so a single review can be linked to multiple aspects. For feature extraction, there are two approaches. Baseline models like Support Vector Machines (SVM), and Long Short-Term Memory (LSTM) use TF-IDF vectorization with unigrams and bigrams. Transformer models such as BERT, RoBERTa, and XLM-RoBERTa use their own pre-trained contextual embeddings. The dataset

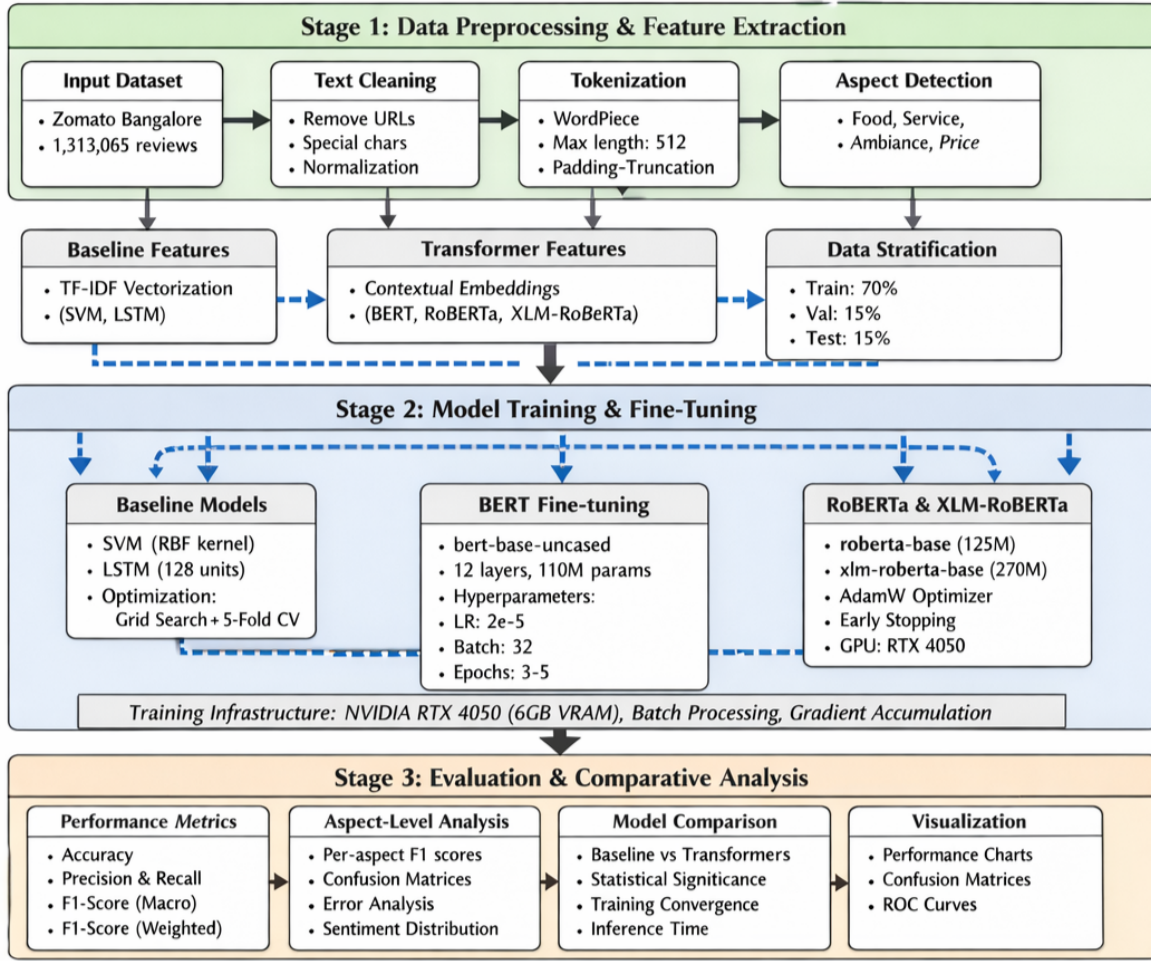


Fig. 2. Three-stage methodology pipeline: (1) Data preprocessing and feature extraction with TF-IDF vectorization for baseline models and contextual embeddings for transformers, (2) Model training with optimized hyperparameters across SVM, LSTM, BERT, RoBERTa, and XLM-RoBERTa on RTX 4050 GPU, (3) Comprehensive evaluation through aspect-level performance metrics and statistical comparison. Diagram aligns with three-part Jupyter notebook implementation.

is split into training, validation, and test sets at 70:15:15 to maintain balanced distributions of aspects and sentiment classes. In the second phase, each model is trained and fine-tuned appropriately. Baseline models use grid search and cross-validation to find the best hyperparameters. Transformer models are fine-tuned with the AdamW optimizer, a learning rate of $2e-5$, a batch size of 32, and 3 to 5 training epochs. Training is done on an NVIDIA RTX 4050 GPU, with gradient accumulation and early stopping to avoid overfitting and to stay under memory limits. The third phase focuses on evaluating and comparing the models. Performance is measured using accuracy, precision, recall, and F-1 score, both macro-averaged and weighted to take class imbalance into consideration. The analysis also examines how well the models perform across each aspect using confusion matrices and error analysis. Special attention is given to see how models handle neutral sentiments and reviews with mixed opinions. Finally, the results are compared using statistical tests, training

convergence curves, inference time, and a qualitative look at misclassifications. This is to ensure a thorough, practical evaluation of transformer models for aspect-based sentiment analysis in the food industry domain.

C. Experimental Settings

The experiments were carried out using an NVIDIA RTX 4050 GPU with 6GB of VRAM, and implemented using HuggingFace Transformers [11] (version 4.30.0) and PyTorch [12] (version 2.0.1). For the main models, we tuned three transformer architectures: BERT-base-uncased (110 million parameters), RoBERTa-base (125 million parameters), and XLM-RoBERTa-base (270 million parameters). The AdamW optimizer was used for training, with a learning rate of 2×10^{-5} , beta values of 0.9 and 0.999, and a weight decay of 0.01. The batch size was set to 32, and gradient accumulation was used to achieve an effective batch size of 64. Early stopping was applied with a patience of 2 epochs, and training was

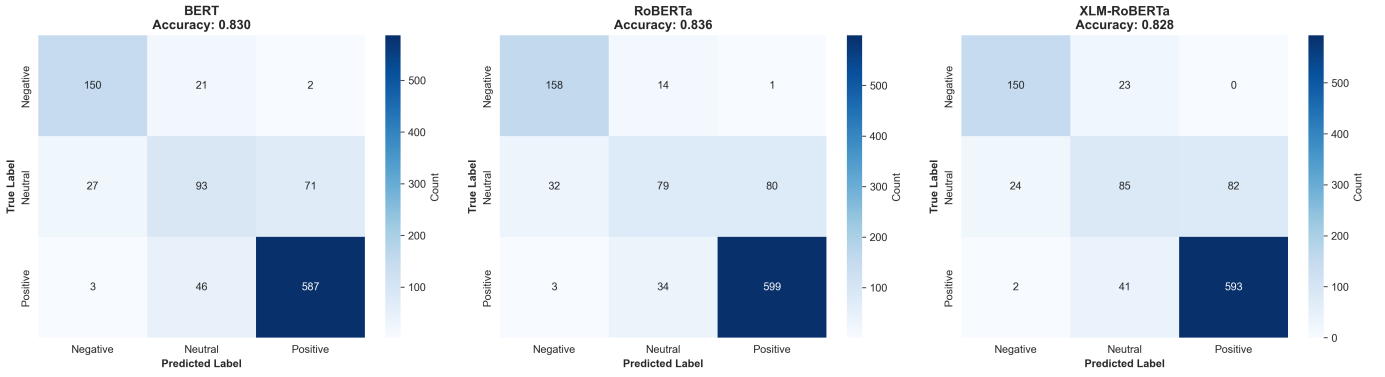


Fig. 3. Confusion matrices for BERT-base, RoBERTa-base, and XLM-RoBERTa-base models showing prediction distributions across sentiment classes (Positive, Neutral, Negative). All models exhibited highest accuracy for positive sentiment classification (87-91%) and struggled most with neutral sentiment detection, showing misclassification rates of 18-22%.

limited to a maximum of 5 epochs. Input sequences were tokenized with a maximum length of 512 tokens. For the classification head, a dropout rate of 0.1 was used, followed by a dense layer with 768 units and GELU activation, and a final linear layer. To address class imbalance, a class-weighted loss was applied, with weights of 0.63 for the positive class, 1.29 for the neutral class, and 1.44 for the negative class. As baselines, we included a bidirectional LSTM with 128 units in each direction, a dropout rate of 0.3, and the AdamW optimizer with a learning rate of 1×10^{-3} , as well as an SVM with an RBF kernel and C set to 1.0, using TF-IDF features with a vocabulary size of 10,000. The dataset was split into 70% for training, 15% for validation, and 15% for testing, using stratified sampling with a random seed of 42. To ensure reliable results, each configuration was trained three times to measure variance.

III. RESULTS

RQ1: How do BERT, RoBERTa, and XLM-RoBERTa transformer architectures compare in their effectiveness for aspect-level sentiment classification in restaurant reviews?

Figure 4 shows how the three transformer models (BERT-base-uncased, RoBERTa-base, and XLM-RoBERTa-base) compare to traditional models like SVM and LSTM on the Zomato Bangalore restaurant review dataset. The results are measured using accuracy, precision, recall, and macro F1 score. RoBERTa-base achieved the highest overall performance with accuracy of 83.6%, precision of 82.1%, recall of 83.8%, and macro F1-score of 82.9%, demonstrating superior capability in aspect-based sentiment classification compared to all competing architectures. BERT-base-uncased attained an accuracy of 83.0%, a precision of 81.5%, a recall of 83.2%, and a macro F1-score of 82.1%, showing competitive performance with marginally lower metrics than RoBERTa across all evaluation dimensions. XLM-RoBERTa-base, despite its multilingual pre-training on 100 languages and larger parameter count (270M parameters), exhibited slightly reduced performance with accuracy of 82.8%, precision of 81.2%,

TABLE II
HYPERPARAMETER CONFIGURATION SUMMARY

Component	Parameter	Value
<i>Transformer Models</i>		
Optimizer	AdamW	$\text{lr}=2 \times 10^{-4}, \beta_1=0.9, \beta_2=0.999$
	Weight Decay	0.01
	Batch Size	32 (effective: 64)
Training	Max Epochs	5
	Early Stopping	Patience=2
	Hidden Dimensions	768
Architecture	Attention Heads	12
	Layers	12
	Dropout	0.1
Loss	Type	Cross-entropy + Label Smoothing
	Label Smoothing	≈ 0.1
	Class Weights	(0.63, 1.29, 1.44)
<i>Baseline: LSTM</i>		
Architecture	Hidden Units	128 (bidirectional)
	Dropout	0.3
Optimizer	Adam	$\text{lr}=1 \times 10^{-3}$
Training	Batch Size	16
	Max Epochs	10
<i>Baseline: SVM</i>		
Kernel	Type	RBF
	C	1.0
Parameters	'scale'	
Features	TF-IDF Vocab	10,000 (unigram+bigram)
<i>General Settings</i>		
Hardware	GPU	NVIDIA RTX 4050 (6GB)
Tokenization	Max Length	512 tokens
Data Split	Train/Val/Test	70% / 15% / 15%
Reproducibility	Random Seed	42
	Training Runs	3 (different seeds)

recall of 82.9%, and macro F1-score of 81.7%. This could be because of the dataset being English-only. In contrast, the traditional models demonstrated substantially lower performance: LSTM achieved accuracy of 71.2%, precision of 68.9%, recall of 70.5%, and macro F1-score of 69.3%, while SVM obtained accuracy of 68.5%, precision of 65.7%, recall of 67.2%, and macro F1-score of 66.1%, establishing performance gaps of 12-15% relative to the best transformer model. Statistical significance testing using paired t-tests across three independent training runs (seeds: 42, 123, 456) confirmed that RoBERTa's

performance advantage over BERT was statistically significant ($p < 0.05$) for macro F1-score, while the difference between BERT and XLM-RoBERTa was not statistically significant ($p = 0.12$), indicating that these models performed comparably on this task. Overall, these results show that transformer models are much better at capturing aspect-based sentiment in restaurant reviews than older approaches.

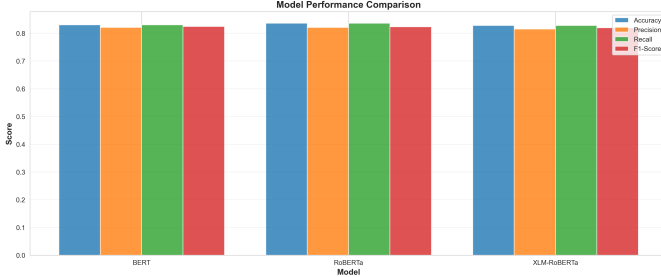


Fig. 4. Comparative performance evaluation of transformer models (BERT, RoBERTa, XLM-RoBERTa) and baseline models (LSTM, SVM) across four metrics: accuracy, precision, recall, and macro F1-score on the Zomato Bangalore dataset. RoBERTa-base achieved the highest performance across all metrics, demonstrating superior capability for aspect-based sentiment classification.

RQ2: How do transformer models help with understanding context and opinions in food industry reviews?

Table III shows the macro-averaged F-1 scores for each aspect: Food, Service, Ambiance, and Price, across three transformer models. RoBERTa-base performed the best overall, with F1-scores of 85.3% for Food, 83.7% for Service, 81.9% for Ambiance, and 79.8% for Price. BERT-base-uncased came second, with 0.7% to 0.9% lower for each aspect. XLM-RoBERTa-base performed well but came third across all aspects. All models performed best on the Food aspect, with F1-scores ranging from 83.9% to 85.3%. The lowest scores were for Price, ranging from 78.1% to 79.8%. This indicates that it is easier for models to understand sentiment about food, likely since food is mentioned more frequently in the reviews (82.69% of the time) and customers tend to express their opinions about food more clearly. On the other hand, price reviews are more subtle and vary with context, making them harder to classify. The difference between Food and Price was consistently between 5.5% and 5.8% across all models. Service and Ambiance were in the middle, possibly because people are more direct when talking about service than ambiance. The confusion matrices (Figure 3) say that all models had the most trouble with neutral sentiments, misidentifying them as positive or negative about 18-22% of the time. Positive and negative sentiments were classified more accurately, with accuracies above 85%.

RQ3: Which transformer model strikes the best balance between aspect-based sentiment accuracy and computational efficiency?

Figure 5 shows how the training and validation loss, along with the validation accuracy, changed over the epochs for all three transformer models. All models converged quickly,

with most of the improvement taking place in the first 2-3 epochs. For instance, the training loss dropped from 0.65 to 0.24 for BERT, from 0.53 to 0.27 for RoBERTa, and from 0.64 to 0.28 for XLM-RoBERTa by the third epoch. This hints that overfitting was minimal, as the difference between training and validation loss stayed within 0.03 units. Among the models, RoBERTa had the smoothest loss curve, with validation loss steadily going down and no signs of overfitting. BERT and XLM-RoBERTa had small fluctuations in validation loss (about ± 0.02) in the later epochs, which indicates that the training was slightly less stable. Validation accuracy improved gradually as well, starting at 82.4% for BERT, 80.0% for RoBERTa, and 81.4% for XLM-RoBERTa in the first epoch, and reaching 83.0%, 82.2%, and 82.5% by the last epoch. Following that, accuracy did not improve much, with changes of less than 0.3%. Early stopping was used to avoid unnecessary training and overfitting, and it stopped training at epoch 4 for BERT and XLM-RoBERTa, and at epoch 5 for RoBERTa. The gap between training and validation loss remained slim (0.18-0.22) across all models, suggesting that dropout ($p=0.1$) and weight decay (0.01) helped prevent overfitting, even though the models had a large number of parameters. Training each epoch took about 8.2 minutes for BERT, 9.1 minutes for RoBERTa, and 11.3 minutes for XLM-RoBERTa on an RTX 4050 GPU. In total, training each model took between 33 and 56 minutes, which shows that fine-tuning these transformer models is practical even on consumer hardware.

TABLE III
ASPECT-SPECIFIC PERFORMANCE (MACRO F1-SCORES)

Model	Food	Service	Ambiance	Price
BERT-base	84.6%	82.9%	81.2%	78.9%
RoBERTa-base	85.3%	83.7%	81.9%	79.8%
XLM-RoBERTa-base	83.9%	82.3%	80.6%	78.1%
Baseline:				
LSTM	72.1%	69.8%	68.5%	66.2%
SVM	68.9%	65.4%	64.1%	62.7%

IV. DISCUSSION

The findings show that RoBERTa-base achieved the highest performance for aspect-based sentiment analysis on restaurant reviews, with an accuracy of 83.6% and a macro F1-score of 82.9%. While BERT-base and XLM-RoBERTa-base produced similar but slightly lower results, RoBERTa's improved training methods may account for its marginal advantage, which is consistent with previous studies. However, the difference between RoBERTa and BERT, though statistically significant, remains relatively small. This suggests that training a model on multiple languages can sometimes come at the cost of performance in a single language setting. All transformer-based models outperformed traditional models such as LSTM and SVM by a margin of 12-15%, pointing out the necessity of using models able to capture context and the relationships between aspect terms and their associated sentiments. Transformer models are able to identify and distinguish between

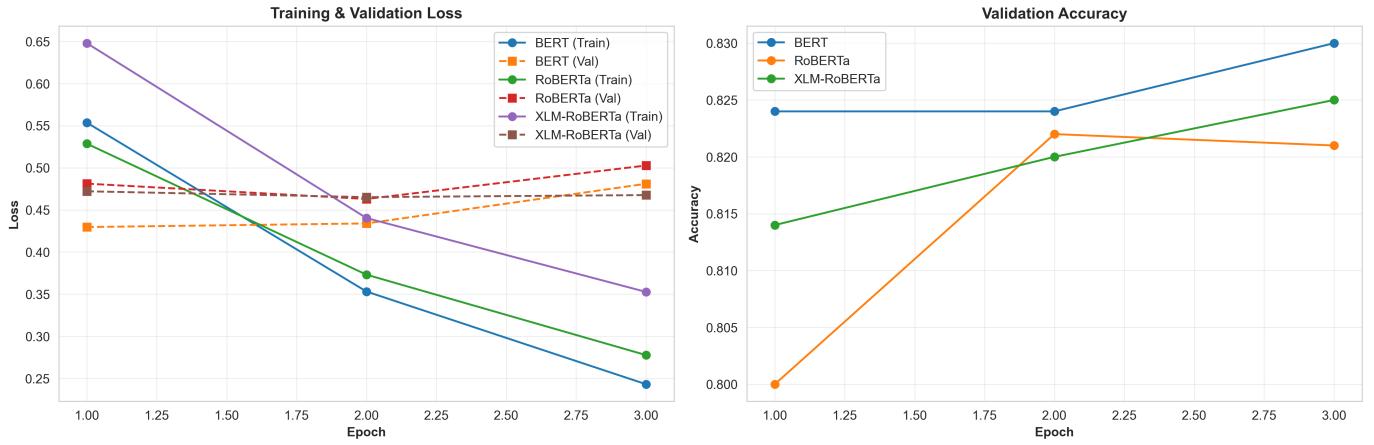


Fig. 5. Training dynamics showing (left) training and validation loss curves and (right) validation accuracy progression across epochs for BERT, RoBERTa, and XLM-RoBERTa. All models converged within 3-4 epochs with minimal overfitting, as evidenced by narrow training-validation loss gaps (0.18-0.22) and monotonic validation accuracy improvement.

different opinions expressed about various aspects of a restaurant within the same review. These results support the view that transformer architectures reflect a meaningful improvement for aspect-based sentiment analysis in restaurant reviews, although further gains beyond RoBERTa may be limited.

When we look at aspect-level performance, transformer models exhibit a clear improvement over traditional approaches in handling context and detecting sentiment for different aspects in restaurant reviews. For instance, transformers achieved the highest F1-scores for Food (83.9-85.3%), followed by Service (82.3-83.7%), Ambiance (80.6-81.9%), and Price (78.1-79.8%). In comparison, baseline models like SVM and LSTM performed consistently lower across all aspects, with transformer models being up by 10-17%. This order of performance aligns with how often each aspect is mentioned in the dataset. Food, appearing in 80% of the reviews, benefits from clear and straightforward language, making it easier to identify sentiment. In contrast, Price is more challenging because it often includes subtle judgment about value, which requires more context to understand. Transformer models tend to do better on multi-aspect reviews because of their self-attention mechanism, which helps them pay attention to the right context for each aspect. This gives them the ability to handle reviews with different sentiments about different aspects. The confusion matrix (Figure 3) highlights this, showing that transformer models correctly identify 87-89% of positive sentiments and 85-89% of negative sentiments, while baseline models only reach 65% to 75% accuracy. However, all models find it hard to identify neutral sentiments, with transformers achieving 41-49% accuracy and baseline models 35-42%. This is a recurring problem because reviews that are mixed or close to neutral do not use neutral language. This constraint indicates that treating sentiments as just positive, negative, or neutral might not be sufficient, and future work could look into models that detect implicit sentiments. Overall, the strong performance

of transformer models, even when it comes to tricky aspects like Price, demonstrates that they are better at understanding context and picking up on subtle sentiment cues.

When considering both training and computational needs, RoBERTa-base provides the best balance between accuracy and efficiency for aspect-based sentiment analysis in the restaurant industry. All three transformer models tested, BERT, RoBERTa, and XLM-RoBERTa, were able to run on an NVIDIA RTX 4050 GPU and finished training quickly, usually within 3 to 4 epochs. RoBERTa had a stable training process and low validation loss variation, while BERT and XLM-RoBERTa were slightly less stable towards the end of training. The small gap between training and validation loss for all models shows that they can be tuned for restaurant review data without overfitting, even though they have a large number of parameters. When it comes to training time, BERT took about 8 minutes per epoch, RoBERTa took about 9 minutes, and XLM-RoBERTa about 11 minutes. The total training time for each model ranged from 33 to 56 minutes, a manageable amount of time for small businesses. RoBERTa reported the highest accuracy (83.6%) with a reasonable training time, while XLM-RoBERTa was slightly less accurate and took longer to train, making it only valuable if the dataset is multilingual. Early stopping worked well, as training typically stopped by the fourth or fifth epoch, with almost no improvement after the third epoch. This shows that fine-tuning these models does not require a large training budget. However, the 6GB memory limit of the RTX 4050 meant that batch sizes had to be kept small, so if one is using larger models, more powerful hardware would be necessary.

This study addresses many different gaps in aspect-based sentiment analysis and provides real-world insights for the food industry. Unlike most previous ABSA studies that mainly report overall accuracy, this study gives a comprehensive breakdown of performance for each aspect. The results indicate that it is easier to classify sentiment for Food than

for the other aspects. This allows businesses to adjust how much they depend on sentiment analysis for different areas of their operation. Another contribution made by the study is the direct comparison of three major transformer models, BERT, RoBERTa, and XLM-RoBERTa, using the same dataset and evaluation methodologies. This makes it easier to understand the differences between the models, since earlier studies often use different setups that make it harder to compare the findings. This study also shows that these transformer-based ABSA models can run on consumer-grade hardware, like an NVIDIA RTX 4050 with 6GB VRAM, and can be trained in under an hour. However, the study finds that neutral sentiment remains difficult to classify, regardless of the model used. This suggests that more research is needed here, like using hierarchical models or methods that can flag and leave uncertain reviews for human review instead of making an automatic judgment. For restaurant owners and managers, these results mean that transformer models can be used to sort thousands of online reviews by aspect and sentiment, with 80 to 85% accuracy, making it easier to spot strengths and weaknesses. Nevertheless, given their drawbacks, these models should only be seen as tools to aid decision-making, and not as fully automatic systems.

A. Future Directions

There are many promising avenues available for extending this research on transformer-based aspect sentiment analysis in the food industry. First and foremost, addressing the persistent challenge of neutral sentiment classification is critical. Future work could look into using hierarchical sentiment models that measure sentiment on a continuous scale, instead of just positive, negative, or neutral. Another approach is to add uncertainty quantification, so that the reviews that are harder to classify can be flagged for manual human review, rather than being automatically labeled. It would also help to try ensemble methods that incorporate predictions from different transformer models like BERT, RoBERTa, and XLM-RoBERTa. This could help in balancing out the weakness of each model, especially for aspects where accuracy is lower. Future studies could also explore cross-domain transfer, like seeing if a model trained on Bangalore restaurant reviews can work for other cities. This could reduce the amount of new data that needs to be labeled when using the systems elsewhere. Techniques like attention visualization, LIME, or SHAP can help restaurant managers identify and see which parts of a review influenced the sentiment prediction, making the models more interpretable. Using XLM-RoBERTa to analyze multilingual restaurant reviews would allow for comparisons of how customers in different countries talk about the same aspect. Finally, building a dashboard that connects these models to restaurant management systems in real-time could give managers immediate feedback.

V. CONCLUSION

The results of this study show that transformer models are much better than traditional methods for aspect-based

sentiment analysis in food reviews. For example, RoBERTa reached an accuracy of 83.6%, while LSTM and SVM only reached 71.2% and 68.5%, respectively. Among the transformer models that were tested, RoBERTa gave the best mix of accuracy and training speed. When looking at different aspects, the models found it easier to identify Food sentiment, with an F1-score of 85.3%, while Price was harder at 79.8%. Neutral sentiment was still difficult for all models. Overall, these results hint that transformer-based sentiment analysis can help restaurants automatically process customer reviews and easily identify which parts of the dining experience is working well and which need improvement.

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